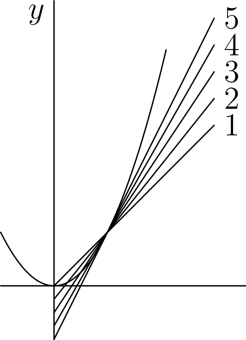

Line 5 is tangent line with slope 2.

Slope of Line 4 $= \frac{f(1)-f(3/4)}{1-\frac{3}{4}} = 7/4$

Slope of Line 3 $= \frac{f(1)-f(1/2)}{1-\frac{1}{2}} = 3/2$

Slope of Line 2 $= \frac{f(1)-f(1/4)}{1-\frac{1}{4}} = 5/4$

Slope of Line 1 $= \frac{f(1)-f(0)}{1-0} = 1$